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VLSI 71

Date :

(3) (i) $x(t) = e^{-t}u(t)$

$y(t) = e^{-2t}\cos(t)\,u(t)$

$X(s) = \frac{1}{s+1}$, $Y(s) = \frac{s+2}{(s+2)^2+1}$

$H(s) = \frac{Y}{X} = \frac{s+2}{(s+2)^2+1} \times (s+1)$

$= \frac{s^2+3s+2}{s^2+4s+5}$

$= \frac{s^2+4s+5-(s+3)}{s^2+4s+5}$

$= 1 - \frac{s+3}{s^2+4s+5}$

$\mathcal{L}^{-1}\left(1 - \frac{s+3}{s^2+4s+5}\right) = \delta(t) - \mathcal{L}^{-1}\left(\frac{s+2}{(s+2)^2+1} + \frac{1}{(s+2)^2+1}\right)$

$= \delta(t) - e^{-2t}u(t)(\cos t + \sin t)$

(ii) $x(t) = e^{-2t}u(t)$

$y(t) = -2e^{-t}u(t) + 2e^{-3t}u(t)$

$X(s) = \frac{1}{s+2}$, $X(s) = \frac{-2}{s+1} + \frac{2}{s+3}$

$$H(s) = \frac{\frac{2}{s+1} + \frac{2}{s+3}}{\frac{1}{s+2}}$$

$$= -\frac{2(s+2)}{(s+1)} + \frac{2(s+2)}{s+3}$$

$$= -2(s+2)\left(\frac{1}{s+3} - \frac{1}{s+1}\right)$$

$$\boxed{H(s) = \frac{-4(s+2)}{(s+3)(s+1)}}$$

$$\frac{s+2}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3}$$

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$$S+2 = A(S+3) + B(S+1)$$

$$= (A+B)S + 3A+B$$

$$A+B=1$$
$$3A+B=2 \Rightarrow A = 1/2$$
$$B = 1/2$$

$$H(S) = -4 \times \frac{1}{2}\left(\frac{1}{S+1} + \frac{1}{S+3}\right)$$

$$= -2\left(\frac{1}{S+1} + \frac{1}{S+3}\right)$$

$$\boxed{h(t) = -2\left(e^{-t} + e^{-3t}\right)u(t)}$$

(6) (a) $X(jw) = \frac{jw}{(2+jw)^2} = jw \times \frac{1}{(2+jw)^2}$

using Prop & table

$$x(t) = te^{-at}u(t) \xleftrightarrow{FT} X(jw) = \frac{1}{(a+jw)^2}$$

$$\frac{d}{dt}x(t) \xleftrightarrow{FT} jwX(jw)$$

$a=2$

$$\Rightarrow x(t) = \frac{d}{dt}\left(te^{-2t}u(t)\right)$$

$$= e^{-2t}u(t) - 2te^{-2t}u(t)$$

$$\boxed{x(t) = (1-2t)e^{-2t}u(t)}$$

(b) $X(jw) = \frac{4\sin(2w-4)}{2w-4} - \frac{4\sin(2w+4)}{2w+4}$

$$= \frac{2\sin(2(w-2))}{w-2} - \frac{2\sin(2(w+2))}{w+2}$$

From table

$$x(t) = \begin{cases} 1, & |t| \le T_0 \\ 0 & |t| \ge T_0 \end{cases} \longleftrightarrow X(jw) = \frac{2\sin(wT_0)}{w}$$

$$e^{j\gamma t}x(t) \xleftrightarrow{FT} X(j(w-\gamma))$$

$T_0 = 2$

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Let $g(t) = \begin{cases} 1 & |t| < 2 \\ 0 & |t| > 2 \end{cases}$

$G(jw) = \frac{2\sin(2w)}{w}$

Applying Shifting

$x(t) = g(t)e^{j2t} - g(t)e^{j(-2)t}$

$= g(t)(e^{j2t} - e^{-2jt})$

$= g(t)\,2j\sin(2t)$

$$x(t) = \begin{cases} 2j\sin(2t), & |t| \le 2 \\ 0 & t > 2 \end{cases}$$

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(7) (a) $x[n] = \left(\frac{1}{3}\right)^n u[n+2]$

using $x[n] = \alpha^n u[n] \longleftrightarrow X(\Omega) = \frac{1}{1-\alpha e^{-j\Omega}}$

$\alpha = 1/3$

$$\left(\frac{1}{3}\right)^n u[n] \xleftrightarrow{DTFT} \frac{1}{1-\frac{1}{3}e^{-j\Omega}}$$

using $x[n-n_0] \xleftrightarrow{DTFT} e^{-j\Omega n_0} X(e^{j\Omega})$

$$x[n] = \left(\frac{1}{3}\right)^{(n+2)-2} u[n+2]$$

$$= \left(\frac{1}{3}\right)^{n+2} \times \left(\frac{1}{3}\right)^{-2} u[n+2]$$

$$= \left(\frac{1}{3}\right)^{n+2} \times u[n+2] \times 9$$

$$X(e^{j\Omega}) = 9 \times e^{j\Omega \times 2} \times \frac{1}{1-\frac{1}{3}e^{-j\Omega}}$$

$$\boxed{X(e^{j\Omega}) = \frac{9e^{2j\Omega}}{1-\frac{1}{3}e^{-j\Omega}}}$$

(b) $x[n] = \left(\cos\left(\frac{\pi n}{4}\right)\right)\left(\frac{1}{2}\right)^n u[n-2]$

$$x[n] = \frac{e^{jn\pi/4} + e^{-jn\pi/4}}{2} \times \left(\frac{1}{2}\right)^n u[n-2]$$

$$= \frac{e^{jn\pi/4}}{2} \times \left(\frac{1}{2}\right)^n u[n-2] + \frac{e^{-jn\pi/4}}{2}\left(\frac{1}{2}\right)^n u[n-2]$$

$$= \frac{e^{jn\pi/4}}{2}\left(\frac{1}{2}\right)^{(n-2)+2} u[n-2] + \frac{e^{-jn\pi/4}}{2}\left(\frac{1}{2}\right)^{(n-2)+2} u[n-2]$$

$$= \frac{e^{jn\pi/4}}{8} \times \left(\frac{1}{2}\right)^{n-2} u[n-2] + \frac{e^{-jn\pi/4}}{8}\left(\frac{1}{2}\right)^{n-2} u[n-2]$$

Using $x[n] = \alpha^n u[n] \longleftrightarrow X(\Omega) = \frac{1}{1-\alpha e^{-j\Omega}}$

$$(1/2)^n u[n] \xleftrightarrow{DTFT} \frac{1}{1-\frac{1}{2}e^{-j\Omega}}$$

$$\left(\frac{1}{2}\right)^{n-2} u[n-2] \xleftrightarrow{DTFT} e^{-j\Omega\times 2}\,\frac{1}{1-\frac{1}{2}e^{-j\Omega}}$$

$$u[n-2]\,\frac{1}{8}\left(\frac{1}{2}\right)^{n-2} \xleftrightarrow{DTFT} \frac{1}{8}e^{-j\Omega 2} \times \frac{1}{1-\frac{1}{2}e^{-j\Omega}}$$

$$e^{j\pi/4 n} \times \frac{1}{8}\left(\frac{1}{2}\right)^{n-2} u[n-2] \xleftrightarrow{DTFT}$$

$$\frac{1}{8}e^{-2j(\Omega-\pi/4)}\,\frac{1}{1-\frac{1}{2}e^{-j(\Omega-\pi/4)}}$$

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$$e^{-j n \pi/4} \frac{1}{8}\left(\frac{1}{2}\right)^{n-2} u[n-2] \xrightarrow{DTFT} \frac{1}{8} \cdot e^{-j2(\Omega+\frac{\pi}{4})} \cdot \frac{1}{1-\frac{1}{2}e^{-j(\Omega+\pi/4)}}$$

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$$e^{j n \frac{\pi}{4}} \cdot \left(\frac{1}{8}\right)\left(\frac{1}{2}\right)^{n-2} \cdot u[n-2] + e^{-j n \frac{\pi}{4}} \frac{1}{8}\left(\frac{1}{2}\right)^{n-2} u[n-2]$$

$$\xrightarrow{} \frac{1}{8} e^{-2j(\Omega-\pi/4)} \frac{1}{1-\frac{1}{2}e^{-j(\Omega-\pi/4)}} + \frac{1}{8} e^{-2j(\Omega+\pi/4)} \times \frac{1}{1-\frac{1}{2}e^{-j(\Omega+\pi/4)}}$$

$$x[n] \xrightarrow{DTFT} \frac{1}{8} e^{-2j(\Omega-\frac{\pi}{4})} \frac{1}{1-\frac{1}{2}e^{-j(\Omega-\pi/4)}} + \frac{1}{8} e^{-2j(\Omega+\frac{\pi}{4})} \frac{1}{1-\frac{1}{2}e^{-j(\Omega+\pi/4)}}$$

$$\therefore X(e^{j\Omega}) = \frac{1}{8} e^{-2j\Omega}\left(\frac{e^{j\pi/2}}{1-\frac{1}{2}e^{-j(\Omega-\pi/4)}} + \frac{e^{-j\pi/2}}{1-\frac{1}{2}e^{-j(\Omega+\pi/4)}}\right)$$

$$\boxed{X(e^{j\Omega}) = \frac{j}{8} e^{-2j\Omega}\left(\frac{1}{1-\frac{1}{2}e^{-j(\Omega-\pi/4)}} - \frac{1}{1-\frac{1}{2}e^{-j(\Omega+\frac{\pi}{4})}}\right)}$$

(9) $T = 8 \quad \omega_0 = 2\pi/T = \pi/4$

$$x(t) = 3\cos\left(\frac{\pi}{4}t + \pi/6\right)$$

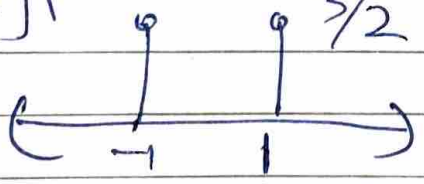
$$= \frac{3}{2}\left(e^{j(\pi/4 t + \pi/6)} + e^{-j(\pi/4 t + \pi/6)}\right)$$

$$= \frac{3}{2}\left(e^{j\pi/4 t} \cdot e^{j\pi/6} + e^{-j\pi/4 t} \cdot e^{-j\pi/6}\right)$$

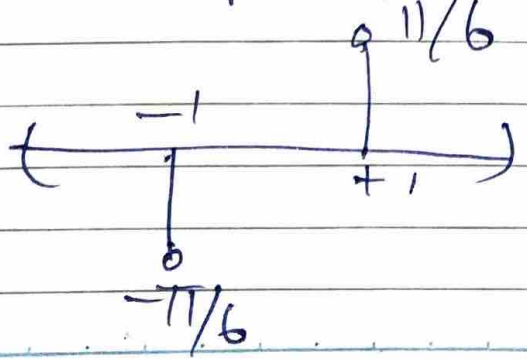
$$= \left(\frac{3}{2}e^{j\pi/6}\right)e^{j\pi/4 t} + \left(\frac{3}{2}e^{-j\pi/6}\right)e^{-j\pi/4 t}$$

$$= \begin{cases} \frac{3}{2}e^{j\pi/6}, & R = 1 \\ \frac{3}{2}e^{-j\pi/6}, & R = -1 \\ 0, & \end{cases}$$

$|X[R]|$



$\arg|X[R]|$



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(10) a) $x[n] = \cos\left(\frac{\pi}{3}n\right) + \sin\left(\frac{\pi}{4}n\right)$

$\frac{2\pi}{\pi/3} = 6 \qquad \frac{2\pi}{\pi/4} = 8$

$LCM(6,8) = 24$

$x[n]$ pd. $= 24$

$\Omega_0 = \frac{2\pi}{24} = \pi/12$

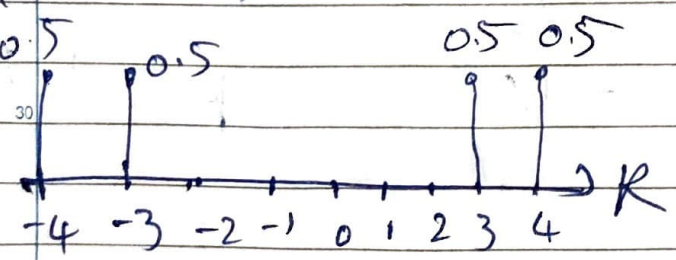
$x[n] = \cos\left(\frac{\pi}{3}n\right) + \sin\left(\frac{\pi}{4}n\right)$

$= \cos(4\Omega_0 n) + \sin(3\Omega_0 n)$

$= \frac{1}{2}e^{j4\Omega_0 n} + \frac{1}{2}e^{-j4\Omega_0 n} + \frac{1}{2j}e^{j3\Omega_0 n} - \frac{1}{2j}e^{-j3\Omega_0 n}$

$x[n] \xleftrightarrow[\pi/12]{DTFS} X[k] = \begin{cases} 1/2, & k=4 \\ 1/2, & k=-4 \\ 1/2j, & k=3 \\ -1/2j, & k=-3 \\ 0, & \end{cases}$

$|X[k]|$



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$\arg|X[R]|$

